

Stability of linear systems

intro:

$$x' = x(2-x-y) \quad \text{eq \#1}$$

$$y' = (1-y)(2+x) \quad \text{eq \#2}$$

• find all critical points

you begin by choosing either " x " or " $(2-x-y)$ " from eq#1 then let it equal zero or $x'=0$ and choose " $(1-y)$ " or " $(2+x)$ " and set that equal to zero so in simpler words you are choosing a term from eq#1 and a term from eq#2

$$\text{let } x=0 \quad \text{and} \quad \text{let } 1-y=0$$

$$\Rightarrow y=1$$

$$(0, 1)$$

• this is one critical point

$$\text{let } x=0 \quad \text{and}$$

$$2+x=0$$

$$x = -2$$

$$0 \neq -2$$

• this is just empty

$$\text{let } 2-x-y=0 \quad \text{and} \quad 1-y=0$$

$$= y=1$$

$$2-x-(1)=0$$

$$x=1$$

• plug this to get value of x

$$(1, 1)$$

$$\text{let } 2-x-y=0$$

and

let

$$2+x=0$$

$$= x=-2$$

$$2-(-2)-y=0$$

$$y=4$$

$$(-2, 4)$$

our critical points are: $(-2, 4)$, $(1, 1)$, $(0, 1)$

notice how we took turns with setting " x " and " $(2-x-y)$ " equal to zero

$$x' = (4-x)(y-x)$$

$$y' = (10-x)(y+x)$$

Find all critical points

$$\begin{aligned} \text{let } 4-x &= 0 \\ \Rightarrow x &= 4 \end{aligned}$$

$$\begin{aligned} \text{and } 10-x &= 0 \\ \Rightarrow x &= 10 \end{aligned}$$

$4 \neq 10$ • this has no critical point

$$\begin{aligned} \text{let } 4-x &= 0 \\ \Rightarrow x &= 4 \end{aligned}$$

$$\begin{aligned} \text{and } y+x &= 0 \\ y+4 &= 0 \\ \Rightarrow y &= -4 \end{aligned}$$

$$(4, -4)$$

$$\begin{aligned} \text{let } y-x &= 0 \\ y &= x \\ \Rightarrow y &= 10 \end{aligned}$$

$$\begin{aligned} \text{and } 10-x &= 0 \\ \Rightarrow x &= 10 \end{aligned}$$

$$(10, 10)$$

$$\begin{aligned} \text{let } y-x &= 0 \\ y &= x \\ \Rightarrow x &= 0 \end{aligned}$$

$$\begin{aligned} \text{and } y+x &= 0 \\ y+y &= 0 \\ 2y &= 0 \\ \Rightarrow y &= 0 \end{aligned}$$

$$(0, 0)$$

$$x' = x^3 - y$$

$$y' = x - y^3$$

let $x^3 - y = 0 \Rightarrow y = x^3$ and let $0 = x - y^3$

now substitute " $y = x^3$ " into " $x - y^3 = 0$ "

$$x - (x^3)^3 = 0$$

$$x - x^9 = 0$$

$$x(1 - x^8) = 0$$

$$x = 0 \quad 1 - x^8 = 0$$

$$x^8 = 1$$

$$x = \pm 1$$

$$\text{if } x = 0 \Rightarrow y = 0^3 = 0$$

$$\text{if } x = -1 \Rightarrow y = -1$$

$$\text{if } x = 1 \Rightarrow y = 1$$

Critical Points $\Rightarrow (0,0), (-1,-1), (1,1)$

Case #1: real distinct eigenvalues

- a) both eigenvalues (λ) are negative and $\gamma^2 - 4\Delta > 0$, $\gamma < 0$, $\Delta > 0$
 $\Rightarrow$ stable node
- b) both eigenvalues (λ) are positive and $\gamma^2 - 4\Delta > 0$, $\gamma > 0$, $\Delta > 0$
 $\Rightarrow$ unstable node
- c) eigenvalues (λ) have opposite signs ($\gamma^2 - 4\Delta > 0$, $\Delta < 0$)
 $\Rightarrow$ saddle points

Case #2: repeated real eigenvalues

- a) $\lambda < 0$ = degenerate stable node
- b) $\lambda > 0$ = degenerate unstable node
- c) $\lambda = 0$ = degenerate stable node

Case #3: complex eigenvalues

- a) $\gamma = 0$ and $\alpha = 0$ $\Rightarrow$ center
- b) $\gamma \neq 0$ and $\alpha \neq 0$ $\Rightarrow$ damped motion
- c) $\alpha < 0$ $\Rightarrow$ stable spiral point

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

trace

$$\gamma = a + d$$

$$\lambda = \frac{\gamma \pm \sqrt{\gamma^2 - 4\Delta}}{2}$$

eigenvalues

$$\Delta = (a)(d) - (c)(b)$$

Classify the critical Point $(0,0)$ for the linear system:

$$A = \begin{pmatrix} 3 & -18 \\ 2 & -9 \end{pmatrix}$$

$$\lambda = \frac{\gamma \pm \sqrt{\gamma^2 - 4\Delta}}{2} \rightarrow \frac{-6 \pm \sqrt{36 - 4(9)}}{2}$$

$$\gamma = 3 + (-9) = -6$$

$$\lambda = -3, -3 \quad \cdot \text{repeated}$$

$$\Delta = (3)(-9) - (2)(-18) = 9$$

degenerate stable node, because $\lambda < 0$

Classify the critical Point $(0,0)$

$$A = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$$

$$\gamma = 2 + 1 = 3$$

$$\lambda = \frac{3 \pm \sqrt{3^2 - 4(-4)}}{2}$$

$$\Delta = (2)(1) - (2)(3) = -4$$

$$\lambda = 4, -1 \quad \cdot \text{different values}$$

Saddle Point because $\Delta < 0$

Classify the critical point $(0,0)$

$$A = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix}$$

$$\chi = \frac{0 \pm \sqrt{0^2 - 4(1)}}{2} = \frac{2i}{2} = i$$

$$\chi = -1 + 1 = 0$$

$$\Delta = (-1)(1) - (-1)(2) = 1$$

$$\lambda = 0 \pm i \quad \bullet \quad \text{Complex}$$

this critical point is a center because $\alpha = 0$